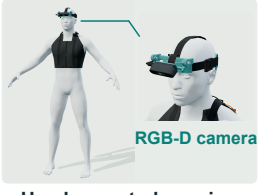
# WEAVE Wearable demonstrations to robot learning

PROPOSED END-TO-END FRAMEWORK

Evaluation failures guide recollection and refinement

## (a) Wearable platform




**Head-mounted sensing**

On-body compute + power

Wearable demonstrations

## (b) Demonstration capture

RGB

Depth

t0 t1 t2

**Time-aligned RGB-D**

Sync + calibration checks

Raw sessions + metadata

## (c) Decomposition & retargeting

**Hand / object poses**

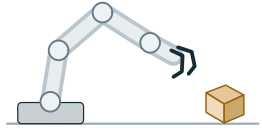
s1 s2 s3

Segment + annotate

Clean + robot retargeting

Robot-ready dataset v1.0

## (d) Robot learning & evaluation



Illustrative robot; platform TBD

Policy training

Simulate + select checkpoint

Real-robot evaluation

## Bench pilot v0

Fixed-camera capture before wearable completion

Same processing path; early annotation and learning

## Parallel learning setup

Public data: learning-pipeline smoke tests

Robot data: independent baseline comparison

### TRACEABLE ARTIFACTS

- **Hardware version** — CAD / BOM / test log
- → **Session identity** — Timestamps / calibration
- → **Dataset version** — Segments / transforms
- → **Experiment identity** — Code / seed / checkpoint

Framework concept, not experimental results. V10 images are CAD references; downstream processing and evaluation are planned.